PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE
EXHAUST ABSORPTION POST
尾气吸收岗位

MTI-CHP-CPM-SOP-014_Rev. 000

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the exhaust absorption post under complete closure.

此程序涵盖操作员对完全关闭下的尾气吸收岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the exhaust absorption post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the exhaust absorption post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对尾气吸收岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定尾气吸收岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP 岗位操作规程
- Desulfurizing Tower (4306-TOW-004). 脱硫塔 (4306-TOW-004)
- Desulfurizing Circulating Pump (4306-PSL-017/018/019). 脱硫循环泵 (4306-PSL-017/018/019)
- Fan (4306-FAN-001/002/003). 风机 (4306-FAN-001/002/003)

4. PPE 个人保护装备



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5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Liquid corrosion; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Poisoning; 7. Object falling from high altitudes; 8. Liquid sputtering; 9. Noise. 1. 机械伤害; 2. 液体腐蚀; 3. 触电; 4. 湿滑的条件，绊倒危险; 5. 窒息; 6. 中毒; 7. 高空坠物; 8. 液体喷溅; 9. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear appropriate labor protection supplies, once the skin contact with hydrochloric acid solution, immediately rinse with a lot of water; 3. Do not contact the electrical switch with wet hands during operation,Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clearclearance or maintenance operations; 6. Wear good labor protection goods, Found smoke or liquid running phenomenon immediately treated and reported; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Wear protective masks and gloves, as well as other protective products as required; 9. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩; 2. 穿戴适宜的劳保用品，一旦皮肤接触到盐酸溶液，立即用大量清水冲洗; 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效; 4. 上下楼梯请扶好扶手; 5. 清仓或维修作业执行审批程序; 6. 穿戴好劳保护品，发现烟气或液体跑冒现象立即处理并汇报; 7. 穿戴好劳保护品，检查各平台的物品堆放安全性; 8. 按要求戴好防护面具和防护手套，以及其它劳保护品; 9. 佩戴好耳塞，保护听力。

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6. Equipment 工具/ 设备

4306-CHY-001	Chimney烟囱
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Desulfurizing Tower (4306-TOW-004) 检查脱硫塔(4306-TOW-004)	1. Remove all sundries in the desulfurization tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water gate valve (50GA) before the circulation tank of the desulfurization tower, fill the desulfurization tower with new water, open the desulfurization cycle pump (4306-PSL-017 / 018 / 019) butterfly valve (350BF, 500BF), and open the desulfurization cycle pump (4306-PSL-017 / 018 / 019); 5. Observe whether the spray and washing condition of the desulfurization tower meets the requirements; 6. Check whether the spray support ring, fog gate plate, screen separator and nozzle assembly are normal, and the spray remover and nozzle are not blocked; 7. Check whether the packing, compactness and height of the packing meet the requirements, and whether the spray dispersion is uniform; 8. Check whether the circulating fluid inlet, access port, sewage outlet, gas inlet and outlet, manhole are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 9. Empty the inspection water in the desulfurization tower.

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№	Stage	Key Points
		1. 清除脱硫塔等设备内的一切杂物； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开脱硫塔循环槽前新水闸阀(50GA)，将脱硫塔注满新水，打开脱硫循环泵(4306-PSL-017/018/019)进出蝶阀(350BF,500BF)，开启脱硫循环泵(4306-PSL-017/018/019)； 5. 观察脱硫塔喷淋洗涤情况是否符合要求； 6. 检查喷淋支座环、除雾栅板、丝网除沫器、喷嘴组件是否正常，除沫器和喷嘴有无堵塞现象； 7. 检查填料排布、密实度和高度是否符合要求，喷淋液散布是否均匀； 8. 检查循环液进口、检修口、排污口、气体进出口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 9. 放空脱硫塔内的检查用水。
3	Inspect Desulfurizing Circulating Pump (4306-PSL-017/018/019) 检查脱硫循环泵 (4306-PSL-017/018/019)	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the pump control cabinet voltmeter, signal lights and other instrument indicators are normal; 7. Check whether there is air in the pump unit, if it should be excluded; 8. Check whether the pump head is under the liquid level of the desulfurization tower (4306-TOW-004); 9. Check that the delivery pump is in normal state, and the outlet butterfly valve (500BF) is temporarily closed after flexible flexibly; 10. Check that the backup pump outlet butterfly valve (500BF) is fully closed. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好； 2. 手转动泵轴3圈，应灵活无阻滞；

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№	Stage	Key Points
		3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3之间； 4. 检查泵的密封状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于脱硫塔(4306-TOW-004)液位下； 9. 检查输送泵处于正常状态，出口蝶阀(500BF)开启灵活后暂时处于关闭状态； 10. 检查备用泵出口蝶阀(500BF)关闭状态。
4	Inspect Fan (4306-FAN-001/002/003) 检查风机(4306-FAN-001/002/003)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2 ~ 3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)。
5	Check the other Equipment and Facilities 检查其它设备设施	1. Chimney (4306-CHY-001) a. Check whether there is debris at the bottom of the chimney, whether there is leakage and ventilation is unobstructed.

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№	Stage	Key Points
		1. 烟囱(4306-CHY-001) a. 检查烟囱底部是否有杂物，是否漏气，通风是否通畅。 2. Valve a. Valves of the desulfurization tower (4306-TOW-004) Check whether the three lime milk ball valve (100BA,100BA,100BA), the new water brake valve (50GA), the injection inlet butterfly valve (400BF) and the spray discharge butterfly valve (65BA) are intact and flexible. b. Fan (4306-FAN-001/002/003) valves Check whether the three inlet butterfly valve (1600BF,1600BF,1600BF), the three outlet butterfly valve (1600BF,1600BF,1600BF) and the three fan liquid discharge ball valve (40BA,40BA,40BA) are intact and flexible. c. Desulphurization cycle pump (4306-PSL-017/018/019) Check whether the four inlet butterfly valve (350BF,350BF,350BF,350BF) and the four outlet butterfly valve (500BF,500BF,500BF,400BF) are intact and flexible. 2. 阀门 a. 脱硫塔(4306-TOW-004)各阀门 检查石灰乳球阀三个(100BA,100BA,100BA)、新水闸阀一个(50GA)、喷淋液入口蝶阀一个(400BF)和喷淋液外排蝶阀一个(65BA)是否完好、灵活。 b. 风机(4306-FAN-001/002/003)各阀门 检查入口蝶阀三个(1600BF,1600BF,1600BF)，出口蝶阀三个(1600BF,1600BF,1600BF)和风机积液外排球阀三个(40BA,40BA,40BA)是否完好、灵活。 c. 脱硫循环泵(4306-PSL-017/018/019)各阀门 检查入口蝶阀四个(350BF,350BF,350BF,350BF)和出口蝶阀四个(500BF,500BF,500BF,400BF)是否完好、灵活。
6	Inspect Materials Level 检查物料料位	1. Check whether the bottom of the desulfurization tower (4306-TOW-004) and reach the specified liquid level height; 2. Check whether each liquid line accumulates fluid； 3. Check whether the lime storage mixing tank (4524-SGT-003) is filled with qualified lime milk.

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№	Stage	Key Points
		1. 检查脱硫塔(4306-TOW-004)底部是否注入合格的石灰乳，并达到规定的液位高度； 2. 检查各液体管路是否积液； 3. 检查石灰储存搅拌槽(4524-SGT-003)是否备满合格的石灰乳。
7	Start-Up Of The Exhaust Absorption Post 尾气吸收岗位开机	2. No-load start should be guaranteed when starting; 3. The driving sequence is along the flue gas. To ensure the protection of the environment, all the equipment needs to be opened later; meanwhile, confirm that the process related to the exhaust gas absorption post has been started and that the feeding conditions are available: At the bottom of the desulfurization tower (4306-TOW-004) and reach the specified liquid level height, and the lime storage mixing tank (4524-SGT-003) is filled with qualified lime milk ⟹ Open the lime milk ball valve (100BA), the injection inlet butterfly valve (400BF) and the spray discharge butterfly valve (350BF), and start the desulfurization tower (4306-TOW-004) ⟹ Open the desulfurization cycle pump (4306-PSL-017/018/019) outlet butterfly valve (500BF) and inlet butterfly valve (350BF), and start the desulfurization cycle pump (4306-PSL-017/018/019) ⟹ Open the fan (4306-FAN-001/002/003) outlet butterfly valve (1600BF) and inlet butterfly valve (1600BF) and start the fan (4306-FAN-001/002/003). 1. 启动时应保证空载启动； 2. 开车顺序为沿烟气走向，为保护环境，从前往后开即顺向开车，各设备都需要延时开启；同时，确认尾气吸收岗位相关的工序已经开车并具备给料条件： 脱硫塔(4306-TOW-004)底部注入合格的石灰乳、并达到规定的液位高度，石灰储存搅拌槽(4524-SGT-003)备满合格的石灰乳 ⟹ 打开石灰乳球阀(100BA)、喷淋液入口蝶阀(400BF)和喷淋液外排蝶阀(350BF)，启动脱硫塔(4306-TOW-004) ⟹ 打开脱硫循环泵(4306-PSL-017/018/019)出口蝶阀(500BF)和入口蝶阀(350BF)，启动脱硫循环泵(4306-PSL-017/018/019) ⟹ 打开风机(4306-FAN-001/002/003)出口蝶阀(1600BF)和入口蝶阀(1600BF)，启动风机(4306-FAN-001/002/003)。
8	The Exhaust Absorption Post Stops	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping;

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№	Stage	Key Points
	尾气吸收岗位停机	b. Stop in the order of protecting the environment, Each equipment needs to stop: Close the inlet butterfly valve (350BF), stop the desulfurization cycle pump (4306-PSL-017/018/019), and close the outlet butterfly valve (500BF)⟹ Close the lime milk ball valve (100BA) and the injection liquid inlet butterfly valve (400BF), stop the desulfurization tower (4306-TOW-004), and the spray liquid discharge butterfly valve (350BF)⟹ Close the inlet butterfly valve (1600BF), stop the fan (4306-FAN-001/002/003), and close the outlet butterfly valve (1600BF)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车; b. 按照保护环境的顺序停车，各设备都需要延时停止： 关闭脱硫循环泵(4306-PSL-017/018/019)入口蝶阀(350BF)，停止脱硫循环泵(4306-PSL-017/018/019)，关闭脱硫循环泵(4306-PSL-017/018/019)出口蝶阀(500BF)⟹关闭石灰乳球阀(100BA)和喷淋液入口蝶阀(400BF)，停止脱硫塔(4306-TOW-004)，关闭喷淋液外排蝶阀(350BF)⟹关闭风机(4306-FAN-001/002/003)入口蝶阀(1600BF)，停止风机(4306-FAN-001/002/003)，关闭风机(4306-FAN-001/002/003)出口蝶阀(1600BF)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency accident occurs, Close fan (4306-FAN-001/002/003), desulfurization cycle pump (4306-PSL-017/018/019), desulfurization tower (4306-TOW-004), After all stop the operation, Check whether the start / stop button of the fan (4306-FAN-001/002/003), desulfurization cycle pump (4306-PSL-017/018/019) are in a stopped state, Close the inlet butterfly valve (1600BF) and the outlet butterfly valve (1600BF) of the shutdown fan (4306-FAN-001/002/003); Close the inlet butterfly valve (350BF) and the outlet butterfly valve (500BF) of the desulfurization cycle pump (4306-PSL-017/018/019); Close the lime milk ball valve (100BA), injection inlet butterfly valve (400BF) and spray discharge butterfly valve (350BF) of the

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№	Stage	Key Points
		desulfurization tower (4306-TOW-004); Calcium chloride solution outlet butterfly valve (600BF); at the same time, The upper level of equipment must stop; In case of an emergency, first shut down the fan (4306-FAN-001/002/003), and then stop according to the parking order; b. After the accident, the equipment returned to normal. If the production is still needed, the operation shall be restarted in the order of driving; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭风机(4306-FAN-001/002/003)、脱硫循环泵(4306-PSL-017/018/019)、脱硫塔(4306-TOW-004)，全部停止运行后，检查风机(4306-FAN-001/002/003)、脱硫循环泵(4306-PSL-017/018/019)的启动/停止按钮是否处于停止状态，关闭关闭风机(4306-FAN-001/002/003)的入口蝶阀(1600BF)和出口蝶阀(1600BF)；关闭脱硫循环泵(4306-PSL-017/018/019)的入口蝶阀(350BF)和出口蝶阀(500BF)；关闭脱硫塔(4306-TOW-004)的石灰乳球阀(100BA)、喷淋液入口蝶阀(400BF)和喷淋液外排蝶阀(350BF)；氯化钙溶液出口蝶阀(600BF)；同时，上一级设备均需停车； 当出现紧急事故首先关停风机(4306-FAN-001/002/003)，然后再按照停车顺序进行停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，无论时间长短再启动时，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
9	Permits &Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Exhaust Absorption Post 尾气吸收岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Exhaust absorption posts in the chloride plant
CP 尾气吸收岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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